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7. LAND CONTAMINATION

7.1 Introduction

7.1.1 This section presents potential land contamination implications associated with the Project. Land contamination implications were assessed in accordance with Clause 3.5.7 and Appendix F of the EIA Study Brief (No. ESB-375/2025).

7.2 Environmental Legislation, Standards and Guidelines

7.2.1 The relevant environmental legislation guidelines and standards on land contamination aspect include the following:-

- Section 3 (Potential Contaminated Land Issues) of Annex 19 "Guidelines for Assessment of Impact on Sites of Cultural Heritage and Other Impacts" of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM).

- Guidance Note for Contaminated Land Assessment and Remediation (Guidance Note)

  This Guidance Note sets out the requirements for the proper assessment and management of potentially contaminated sites such as oil installations (e.g. oil depots, petrol filling stations), gas works, power plants, shipyards / boatyards, chemical manufacturing / processing plants, steel mills / metal workshops, car repairing / dismantling workshops and scrap yards. In addition, this Guidance Note provides guidelines on how site assessments should be conducted and evaluated and suggests practical remedial measures that can be adopted for the remediation of contaminated sites.

- Practice Guide for Investigation and Remediation of Contaminated Land (Practice Guide)

  This guide outlines typical investigation methods and remediation strategies for the range of potential contaminants typically encountered in Hong Kong.

- Guidance Manual for Use of Risk-based Remediation Goals for Contaminated Land Management (Guidance Manual)

  This Guidance Manual introduces the risk-based approach in land contamination assessment and provides details for the comparison of soil and groundwater data under Risk-Based Remediation Goals (RBRGs) for 54 chemicals of concern (COCs) commonly found in Hong Kong. The RBRGs were derived to suit Hong Kong conditions by following the international practice of adopting a risk-based methodology for contaminated land assessment and remediation; and were designed to protect the health of people who could potentially be exposed to land impacted by chemicals under four broad post restoration land use categories. The RBRGs also serve as the remediation targets where remediation is necessary.

7.3 Assessment Methodology

7.3.1 Land contamination assessment was undertaken in accordance with the EIAO-TM, Guidance Note, Practice Guide and Guidance Manual, and the requirements stated in Clause 3.5.7 and Appendix F of the EIA Study Brief.

7.3.2 The following tasks were undertaken to identify any potentially contaminated areas for the purpose of land contamination assessment:

- Site walkover to identify the potentially contaminated sites; and
- Desktop review of site history.

7.3.3 For the desktop review, the following information was reviewed:

- Topographic maps and relevant aerial photographs (Lands Department);
- Hong Kong Geological Survey Maps and previous ground investigation records;
- Relevant EIA reports and Project Profiles
- Records of dangerous goods and chemical spillage/leakage incidents (Fire Services Department, FSD);
- Chemical waste producer register and chemical spillage/leakage incidents (Environmental Protection Department, EPD); and
- Geochemical Atlas of Hong Kong.

7.4 Description of Environment

7.4.1 The Project is located at the Loop, Ma Tso Lung and the rural areas connecting them with the land use comprising fishponds, rural roads (connecting Lok Ma Chau and Ma Tso Lung) and Ma Tso Lung Shun Yee San Tsuen. The Project location is shown in **60722948/A10/FIGURE 2.1**.

7.5 Identification of Potential Land Contamination Concern

Site Walkover

7.5.1 Site walkover was conducted in May 2024, August 2024 and March 2025. Photographs taken during the site walkover are provided in **Appendix 7.1**. A site walkover checklist is provided in **Appendix 7.2**. Based on the findings of the Site walkover, no potential land contamination activity or hotspots were identified within Project boundary or in off-site areas immediately adjacent to the Project Site except for at site MTL-001 along Ma Tso Lung Road.

7.5.2 Site MTL-001 was an open storage for construction company. The site was inaccessible during site visit. The site walkover was conducted through peripheral inspection and drone reconnaissance. The site was an equipment/machinery open storage with potential maintenance and small diesel storage. Equipment and machines (drill rig, pumping, casing/drill rod) for ground investigation were observed on site. Storage of diesel and maintenance of drill rig is anticipated on site. The location of site MTL-001 is provided in **60722948/A10/FIGURE 7.4**.

Review of Historical Land Uses

7.5.3 The development history of the Project area was reviewed using aerial photographs covering the period 1973 to 2022. Selected aerial photographs are presented in **Appendix 7.3**. Historical land use within the Project area is summarised in **Table 7.1**.

7.5.4 Aerial photographs indicate the alignment runs through rural areas with village development. No potential land contamination is identified. Furthermore, until 2013, most of the area lay within the frontier closed area.

7.5.5 However, with reference to the EIA report for Development of Lok Ma Chau Loop (EIA Register No. AEIAR-176/2013), as a result of disposal of contaminated mud within the Loop between 1995-2000; potential land contamination was anticipated. This information is detailed in **Section 7.5.10 - 7.5.13**.

Table 7.1 Historical Land Use within the Project Boundary

Year	Historical Land Uses
1973	The Loop • The area was agricultural land. Between the Loop and Ma Tso Lung • The area was mostly vacant. Near Ma Tso Lung Shun Yee San Tsuen • Village development with agricultural land and rural areas.
1983	The Loop • No information available. Between the Loop and Ma Tso Lung • Fishponds were identified near the Loop. Other areas were mostly vacant with an access road between Lok Ma Chau and Ma Tso Lung. Near Ma Tso Lung Shun Yee San Tsuen • No significant change observed in comparison with land use in the 1973 aerial photograph.
1994	The Loop • No information available. Between the Loop and Ma Tso Lung • No significant change in land use is observed in comparison with the land use from 1983 aerial photo. Lok Ma Chau (Sandy Spur) Operation Base was first observed adjacent to the project boundary. Near Ma Tso Lung Shun Yee San Tsuen • No significant change observed in comparison with land use in the 1983 aerial photograph.
2001	The Loop • No information available. Between the Loop and Ma Tso Lung • No significant change observed in comparison with land use in the 1994 aerial photograph.

Year	Historical Land Uses
	Near Ma Tso Lung Shun Yee San Tsuen • No significant change observed in comparison with land use in the 1994aerial photograph.
2008	The Loop • The area was vacant. Between the Loop and Ma Tso Lung • No significant change observed in comparison with land use in the 2001 aerial photograph. Near Ma Tso Lung Shun Yee San Tsuen • No significant change observed in comparison with land use in the 2001 aerial photograph.
2014	The Loop • No significant change observed in comparison with land use in the 2008 aerial photograph. Between the Loop and Ma Tso Lung • No significant change observed in comparison with land use in the 2008 aerial photograph. Near Ma Tso Lung Shun Yee San Tsuen • No significant change observed in comparison with land use in the 2008 aerial photograph.
2017	The Loop • No significant change observed in comparison with land use in the 2014 aerial photograph. Between the Loop and Ma Tso Lung • No significant change observed in comparison with land use in the 2014 aerial photograph. Near Ma Tso Lung Shun Yee San Tsuen • No significant change observed in comparison with land use in the 2014 aerial photograph.
2022	The Loop • The area was undergoing development. Between the Loop and Ma Tso Lung • No significant change observed in comparison with land use in the 2017 aerial photograph. Near Ma Tso Lung Shun Yee San Tsuen • No significant change observed in comparison with land use in the 2017 aerial photograph.

Noted:
Lands Department aerial photograph for the Loop only available after 2005 (except for 1973). For aerial photo before 2005, the Loop was masked as it was not part of Hong Kong.

Regional Geological Setting

7.5.6 According to the 1:20,000 Hong Kong Geological Survey (HKGS) Solid and Superficial Geology Map Sheet 2 (2022) and Solid Geology Map Sheet 2 (2022), superficial geology is represented by pond deposits, estuarine deposits, marine deposits, alluvium and colluvium. The underlying solid geology includes the Mai Po Member (mainly metasiltstone, metasandstone and graphite-bearing) (Cslm) of the Lok Ma Chau Formation and Tai Shek Mo Member (mainly metasandstone with metaconglomerate and phyllite) (Cslt) of Lok Ma Chau Formation. Medium-grained granite (gm) is mapped at the northwest portion of the Site. The HKGS Solid and Superficial Geology Map is provided in **60722948/A10/FIGURE 7.5**.

7.5.7 **Superficial Geology** - Superficial deposits including fill, pond, marine and estuarine deposits, alluvium, and colluvium are further described as follows.

- Fill is inferred to occur at the developed areas, in particular the former Central Government-Aided Emergency Hospital area. The fill is anticipated to be generally absent in undeveloped areas and natural hillsides within the Project.

- Pond deposits (PD) are anticipated at the location of current and previous ponds mainly located in the vicinity of the central portion of the Site based on information from Map Sheet 2, topographical maps and aerial photographs. PD are anticipated to be underlain by estuarine deposits, marine deposit, alluvium and in-situ materials.

- Intertidal deposits Qhi (ED) are anticipated in places in the vicinity of pond areas to the south of Shenzhen River. ED are anticipated to be overlain by fill or PD and underlain by alluvium and/or in-situ materials.

- Alluvial deposits (including Alluvium Qfa) are anticipated within the majority footprint of the Site and are anticipated to be partially overlain by fill, PD, ED and/or MD.

- Colluvium (COLL) or debris flow deposit Qd/Qcd may be encountered at the footslopes of hillsides at Ping Hang and Ma Tso Lung. COLL is typically underlain by in-situ materials.

7.5.8 **Solid Geology** - The Site area is underlain by complex solid geology with rock types including:

- Lok Ma Chau Formation (Mai Po Member) (Cslm) to the northwestern side of the Ma Tso Lung Fault and southeast of Yuen Tau Shan Fault, i.e., to the northwest of Pang Hang; and also in general area near Lo Wu Range located at the southern portion of the Site. The LMC formation (Mai Po Member) typically consists of meta-sedimentary rocks including metasiltstone, metasandstone and graphite-bearing rocks.

- Lok Ma Chau Formation (Tai Shek Mo Member) (Cslt) generally located to the southeastern side of the Ma Tso Lung Fault, at Ma Tso Lung area where hilly terrain exists. The LMC formation (Tai Shek Mo Member) typically consists of meta-sedimentary rocks including meta-sandstone with metaconglomerate and phyllite.

- Porphyritic coarse-grained granite (undifferentiated) (Ku_gc) is anticipated locally in the northwestern most area of the Site (to the northwest of Yuen Tau Shan

Fault) which mainly comprises coarse-grained granite. Mylonitized granite is anticipated near Yuen Tau Shan Fault.

Review of Previous EIA Study

7.5.9 The following EIA reports and Project Profiles (for permission to apply directly for an Environmental Permit) were reviewed.

- Development of Lok Ma Chau Loop EIA Report (EIA Register No. AEIAR-176/2013)
- Construction of a Secondary Boundary Fence and new sections of Primary Boundary Fence and Boundary Patrol Road EIA Report (EIA Register No. AEIAR-136/2009)
- Project Profile for Installation of the Proposed 132kV Cable Circuits Connecting with Ho To West Substation and Existing 132kV Fanling to Mai Po Cable Circuits (Application No. DIR-284/2021)
- Project Profile for New Boundary Bridge between Lok Ma Chau and Huanggang (Application No. DIR-081/2003)

7.5.10 No potentially contaminated sites were identified in the Construction of a Secondary Boundary Fence and new sections of Primary Boundary Fence and Boundary Patrol Road EIA Report (EIA Register No. AEIAR-136/2009) and the Project Profiles for Installation of the Proposed 132kV Cable Circuits Connecting with Ho To West Substation and Existing 132kV Fanling to Mai Po Cable Circuits (Application No. DIR-284/2021) and New Boundary Bridge between Lok Ma Chau and Huanggang (Application No. DIR-081/2003).

7.5.11 With reference to the EIA report for Development of Lok Ma Chau Loop (EIA Register No. AEIAR-176/2013), potential land contamination was anticipated as a result of disposal of contaminated mud within the Loop (Potential Contaminated Site A) in year 1995 -2000. The depth of disposed mud was anticipated to be approximately 5m (including 1m to 1.5m thickness of a top layer of uncontaminated mud for capping). Site investigation works within the Loop included investigation to a maximum depth of 5m based on the depth of disposed mud. The study area of EIA report for Development of the Loop and the Potential Contaminated Site A is shown in **60722948/A10/FIGURE 7.2**. Previous site investigation locations under the EIA for the Development of the Loop are provided in **60722948/A10/FIGURE 7.2**. Relevant reports from the EIA reports (Appendix 8-2) are extracted and provided in **Appendix 7.4**.

7.5.12 The concentration of the metal “Arsenic (As)” in 6 soil samples collected from 5 boreholes marginally exceeded the Risk-Based Remediation Goals (RBRGs) for Rural Residential and Urban Residential land uses. The 5 boreholes, namely A-S01, A-S01, A-S03, A-S03a and A-S03c1, are not situated within the Project Boundary. However the contaminated areas as identified in the Remediation Action Plan are shown in **60722948/A10/FIGURE 7.2**. Following implementation of the Remediation Action Plan during the construction phase; all contaminated soils will be remediated. No groundwater contamination was detected.

7.5.13 Remediation for arsenic contamination within the Loop was completed by Contract No. YL/2017/03 - Development of the Loop – land decontamination and advance engineering works. Arsenic contaminated soils were treated by cement solidification/stabilisation and backfilled on site.

7.5.14 The previous soil and groundwater test results in the Loop compared against the lower of Industrial or Public Park RBRGs are summarised in **Appendix 7.5**. With reference to **Section 2**, this Project will consist of a depressed/underpass/at grade and viaduct road, footpath, plant room/control room, noise barrier and other ancillary works. The corresponding RBRGs for this project are the "Lower of Industrial or Public Park" as per the Guidance Manual. Although the EIA report reports RBRG exceedance for Arsenic for the Urban Residential and Rural Residential land use scenarios, no exceedance of the lower of Industrial or Public Park RBRGs or soil saturation limits / solubility limits were identified from comparison of previous soil and groundwater testing results. No land contamination associated with mud disposal within the Loop is identified.

Information from Government Departments

7.5.15 The EPD and FSD were contacted to obtain;

- records of any spillage / leakage of chemicals;
- records of Dangerous Goods (DG);
- records of Chemical Waste Producer(s) (CWPs); and
- records of reported incidents within the Project area.

7.5.16 EPD and FSD's replies on the request are presented in **Appendices 7.6** and **7.7**, respectively. The information is summarised below.

Environmental Protection Department

7.5.17 Based on EPD replies dated 13 March 2024, 3 September 2024 and 20 August 2025 regarding requests for chemical spillage/ leakage information, EPD has no record of any chemical spillage / leakage within the Project area. EPD Environmental Compliance Division, Territorial Control Office, Chemical Waste and WEEE (Licensing and Control Support) Section was visited on 8 February 2024 and 12 August 2025 to review CWP records. No CWPs were identified within the Project area.

Fire Services Department

7.5.18 Records of any Dangerous Goods Store(s) and any reported accidents or spillage/leakage within the Project area were requested from FSD. FSD had three letter replies on 12 April 2024, 24 September 2024, and 29 August 2025. No Dangerous Goods store is recorded within the Project area. A total of thirteen (13) individual incidents were reported by FSD with twelve of them being rubbish fire, vegetation fire or No. 1 fire alarm. The approximate location of the incident is provided in **60722948/A10/FIGURE 7.3**.

7.5.19 One of 13 incidents (Event No. FSD10) was chemical spillage in Central Government Aided Emergency Hospital. There was leakage of suspected peroxyacetic acid from 2 nos. of 5-litre plastic containers. It is reported that FSD cleaned up the spilled with absorbent which was sealed in an overpacked drum for EPD disposal. Peroxyacetic

acid is an antimicrobial agent, surface cleaner and sanitiser in hospital. It is a peroxide and readily reacted with (oxidise) other substances. Peroxyacetic acid readily reduced to acetic acid and water and does not persist in the environment. Since the spillage was cleaned up and the subject chemical is readily reacted, land contamination is not anticipated from the reported chemical incident.

Geochemical Atlas of Hong Kong

7.5.20 With reference to the *Preliminary Feasibility Study on Developing the New Territories North*, the *Geochemical Atlas of Hong Kong* published by the Geotechnical Engineering Office of Civil Engineering and Development Department in 1999 and the Hong Kong Environmental Database (HKED) by EPD, natural occurrences of elevated level of arsenic exist over broad areas of the northern New Territories. The arsenic area (>571 mg/kg) as extracted from HKED is provided in **60722948/A10/FIGURE 7.1**. A separate arsenic assessment and remediation plan should be prepared and submitted to EPD for endorsement.

7.6 Prediction and Evaluation of Environmental Impacts

7.6.1 Based on the findings from desktop review (e.g. review of historical aerial photographs and relevant information from EPD and FSD, examination of previous EIA reports and site investigation records) and a site walkover; no land use with potential to give rise to land contamination is identified within Project boundary or at off-site areas immediately adjacent to the Project Site except for Site MTL-001.

7.6.2 Site MTL-001 was a construction company storage yard which is identified as a potentially contaminated site. Possible contaminating activities may have resulted from release of oils and fuels and lubricants from vehicles/plant during storage, maintenance or refuelling as well as storage and use of chemicals and solvents required for maintenance activities and possible waste arisings. A site appraisal report (SAP) for site MTL-001 is provided in **Appendix 7.8**. The location of site MTL-001 is provided in **60722948/A10/FIGURE 7.4**.

7.6.3 Previous Development of the Loop EIA report identified arsenic contaminated soils within the Loop assuming post redevelopment Rural Residential and Urban Residential land uses scenarios. Remediation works for arsenic contamination with the Loop was completed by Contract No. YL/2017/03 - Development of the Loop – land decontamination and advance engineering works. All identified arsenic contaminated soils were treated by cement solidification/stabilisation and backfilled on site.

7.6.4 Site investigation works within the Loop cover a depth of 5m for the previous contaminated mud disposal. A further review of soil and groundwater test results identified no exceedance of the lower of Industrial or Public Park RBRGs and no exceedance of soil saturation limit/solubility limits are identified. No land contamination is identified as a result of previous contaminated mud disposal.

7.6.5 As no land contamination impact arising from the Project is anticipated; no mitigation measures are necessary and no further site investigation for land contamination is required except for Site MTL-001.

7.7 Recommendation of Further Works

7.7.1 The area within the Project boundary will remain in operation and potential exists for a change in land use or operations prior to development. It is recommended to re-appraise the whole area within the Project boundary once the Project Site is resumed and fully accessible. If there are land use changes which pose a potential for land contamination, this information should be reported in Contamination Assessment Plan (CAP) for potentially contaminated sites.

7.7.2 Prior to construction at potentially contaminated sites, the Project Proponent should appoint a consultant to re-appraise the site and update the findings (e.g. locations of hotspots) and sampling and testing requirements presented in the SAP, once the Project Site is resumed and fully accessible. A CAP shall be prepared based on the findings in the SAP alongside incorporating the additional assessment and evaluation findings of the site re-appraisal and the updated sampling and testing strategy should be prepared and submitted to EPD for approval prior to conducting any SI works.

7.7.3 SI works should be implemented according to the CAP. A Contamination Assessment Report (CAR) detailing the findings and, if contaminated soil and/or groundwater is identified, a Remediation Action Plan (RAP) should be prepared and submitted to EPD for approval. Any contaminated soil and groundwater should be treated according to EPD's approved RAP and a Remediation Report (RR) should be submitted to EPD for agreement after completion of the remediation works. No construction or development works should be carried out on potentially contaminated site prior to EPD approval of the CAP, and CAR and/or RR, if necessary.

7.7.4 With reference to the *Preliminary Feasibility Study on Developing the New Territories North*, the *Geochemical Atlas of Hong Kong* published by the Geotechnical Engineering Office of Civil Engineering and Development Department in 1999 and the Hong Kong Environmental Database (HKED) by EPD, natural occurrences of elevated level of arsenic exist over broad areas of the northern New Territories. A separate arsenic assessment and remediation plan should be prepared and submitted to EPD for endorsement.

7.8 Evaluation of Residual Impacts

7.8.1 Referring to Section 8 of the SAP (**Appendix 7.8**), land contamination issues are surmountable. If the recommended further works in **Section 7.7** are followed and contaminated soil and groundwater (if any) is properly treated using appropriate remediation methods and according to EPD's approved RAP. However, any localised soil/groundwater contamination within these sites, if left untreated, could potentially pose health risks to existing or future occupants.

7.8.2 By implementing the recommended further works under this Project, any contaminated soil and groundwater would be identified and treated. The recommended further works would not only minimise the health risk to future occupants arising from exposure to contaminated soil and/or groundwater, it would also provide the opportunity to treat the contaminated soil / groundwater using proven remediation techniques for reuse as useful materials (such as backfilling materials).

7.9 Environmental Monitoring and Audit

7.9.1 Remediation works, if necessary, would be carried out based on the recommended further works outlined in **Section 7.7**. Mitigation measures as recommended in the future EPD approved RAP should be implemented during the remediation works. The

EM&A requirements should be carried out in the form of regular site inspection to ensure the recommended mitigation measures are properly implemented.

7.10 Conclusion

7.10.1 This land contamination assessment assessed the potential for contaminative land use within the Project boundary. Based on the findings of desktop review (e.g. a review of historical aerial photographs and relevant information from EPD and FSD) and a site walkover; no potential land contaminating land use was identified except for Site MTL-001. Site MTL-001 was inaccessible during site visit. The site walkover was conducted through peripheral inspection and drone reconnaissance. The site was a construction company storage yard which is identified as a potentially contaminated site. Possible contaminating activities may have resulted from release of oils and fuels and lubricants from vehicles/plant during storage, maintenance or refuelling as well as storage and use of chemicals and solvents required for maintenance activities and possible waste arisings.

7.10.2 It is recommended to re-appraise the whole area within the Project boundary once the Project Site is resumed and fully accessible prior to development. Further works, including site re-appraisal, SI works as well as submission of a CAP and CAR for EPD's endorsement are recommended to be carried out after the Project site is handed over to the PP for development. If contaminated soil and/or groundwater is identified; remediation should be carried out according to the EPD approved RAP and a RR should be submitted to EPD for agreement after completion of the remediation works. No construction or development works should be carried out on potentially contaminated sites prior to EPD approval of the CAP, and CAR and/or RR, if necessary.

7.10.3 With reference to previous studies, natural occurrences of elevated level of arsenic exist over broad areas of the northern New Territories. A separate arsenic assessment and remediation plan should be prepared and submitted to EPD for endorsement.

7.10.4 With the implementation of the recommended further works, any soil/groundwater contamination would be identified and properly treated prior to the construction works. No insurmountable land contamination impacts to the Project are therefore anticipated.

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